

For issues related to the isotopy conjecture, we refer to [96, Sect. 8.6].

The Lawrence construction is due to Jim Lawrence (surprise), but he never published it. It appears in Billera & Munson [77, Sect. 2] and is also explained in detail (in oriented matroid terms) in [96, Sect. 9.3].

A “ σ -construction” to produce rigid 6-polytopes from planar configurations was given by Sturmfels in [530]. However, we found the arguments in [530] to be incorrect. (Specifically, the claim that the orientation of all the “outer simplices” of a polytope is determined by the combinatorics of the face lattice is only true for simplicial polytopes: for this, consider the cone over a nonrigid polytope, where all simplices are outer; however, the polytopes produced by the σ -construction are not simplicial, and they turn out not to be rigid in general.)

Problems and Exercises

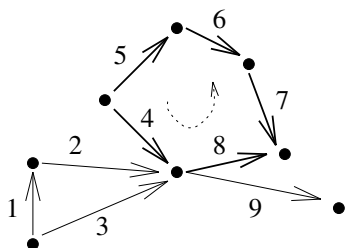
- 6.0 Prove *Radon’s theorem*: given any set V of $d + 2$ points in $\mathbb{R}^d$, we can find disjoint nonempty subsets $V_1, V_2 \subseteq V$ such that $\text{relint}(V_1) \cap \text{relint}(V_2) \neq \emptyset$. Why can we assume that $\text{conv}(V_i)$ are simplices?
- 6.1 Show that if two configurations of n points in $\mathbb{R}^d$ have the same set of minimal affine dependences, then they are affinely isomorphic.
- 6.2 List all the circuits and cocircuits for the hexagon discussed in Section 6.1. How many vectors and covectors are there? (Don’t list them all, there are many.)
- 6.3 Show that the definitions of vectors, circuits, and so on, for the affine and linear cases are consistent: if $V = \begin{pmatrix} \mathbb{1} \\ X \end{pmatrix}$, then

$$\text{a-Dep}(X) = \text{Dep}(V) \quad \text{and} \quad \text{a-Val}(X) = \text{Val}(V),$$

and thus we get the same oriented matroid (the same circuits, cocircuits, etc.) for X and for V .

- 6.4 Let $D = (V, A)$ be a directed graph with arc set $A = \{1, 2, \dots, n\}$.

Define the signed circuits of D to be the sign vectors $u \in \{0, +, -\}^n$ that correspond to circuits in D together with a chosen orientation, as follows: if the arc i is not contained in the circuit, then $u_i = 0$; if it is in the circuit and directed according to the orientation, then $u_i = +$; and if it is directed opposite to the orientation, then $u_i = -$.



$$U = \begin{pmatrix} 0 \\ 0 \\ 0 \\ + \\ - \\ - \\ - \\ + \\ 0 \end{pmatrix}$$

For example, for the digraph drawn here and the oriented circuit marked in it we read off the signed circuit U next to the drawing.

Show that the signed circuits we get that way are from a realizable oriented matroid (as in Definition 6.5), whose cocircuits correspond to the minimal directed cuts in the graph. Interpret the vectors and the covectors of this oriented matroid in terms of the graph.

(Hint: Associate a vector configuration with D . A canonical choice is $v_{ij} = e_i - e_j$ for an arc from the node j to the node i .)

- 6.5 Prove that our two characterizations of *acyclic* vector configurations are equivalent. Prove that the dual of an acyclic configuration is totally cyclic (Corollary 6.16).

Describe a small vector configuration that is neither acyclic nor totally cyclic.

- 6.6 A d -polytope with n vertices is simplicial if and only if every non-empty coface has at least $n - d$ elements. Derive from this a characterization of the (affine) Gale diagrams that represent simplicial polytopes.

- 6.7 Given a Gale diagram, how can one (computationally) enumerate the facets of the corresponding polytope?

- 6.8 Show that the following diagrams represent the four different combinatorial types of 4-polytopes with 6 vertices.



Describe the polytopes.

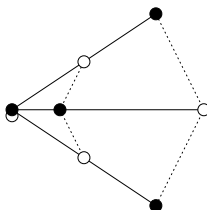
How many facets, and how many edges do they have? Which polytopes are simple or simplicial? Which is $\text{bipyr}(\Delta_3)$? Which is $C_4(6)$?

- 6.9 Describe all the 4-polytopes with 7 vertices. For this, use all the “visualization tools” that we have developed so far:

- Schlegel diagrams
- Gale diagrams
- combinatorial descriptions (vertex-facet matrix)

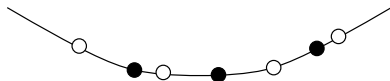
and show how the various types of data correspond to each other.

- 6.10* What is the smallest number of vertices for a nonrational 4-polytopes? (Non-rational 4-polytopes exist by Richter-Gebert's universality theorem, see the Notes for this chapter. The smallest (explicit) example by Richter-Gebert has 33 vertices and ?? facets [459, p. 80]. The minimal number of vertices and facets is not known.)
- 6.11 Analyze the 4-dimensional polytope X^* with 8 facets and 12 vertices whose polar is given by the affine Gale diagram



Show that X^* has a hexagon 2-face whose shape cannot be prescribed. Verify that X^* is combinatorially equivalent to the polytope whose Schlegel diagram is given by Exercise 5.11.

- 6.12 Draw an arbitrary “nice” configuration of black and white points into the plane, and analyze:
- (i) Is this the Gale diagram of a polytope? (If not, add points to get one.)
 - (ii) What is its dimension and its number of facets?
 - (iii) Is it simple or simplicial? Can you describe its facets?
 - (iv) Are there vertices that are not adjacent? Can you compute the graph?
- 6.13 For $n \geq d \geq 2$, consider the moment curve in $\mathbb{R}^{n-d-2}$, and place on it n points, alternating between negative and positive points.



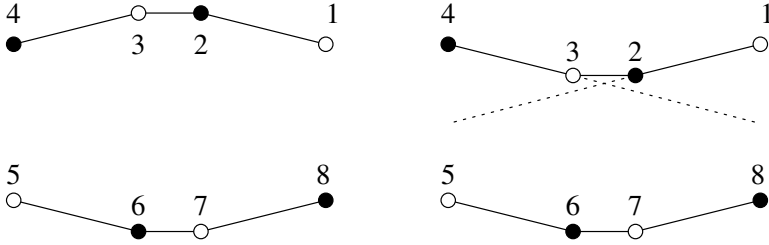
Show that the oriented matroid of this is dual to the oriented matroid of the cyclic polytope $C_d(n)$: so it is the Gale diagram of a polytope that is combinatorially equivalent to $C_d(n)$. Is this really a Gale transform of $C_d(n)$?

- 6.14 Show that the following conditions are equivalent for a $2d$ -dimensional polytope P with n vertices:
- (i) P is neighborly.
 - (ii) Every set of $n - d$ points forms a positive covector.

- (iii) Every circuit of P contains exactly $d + 1$ positive and $d + 1$ negative elements.

Derive the corresponding criteria to detect whether a Gale diagram represents an (even-dimensional) neighborly polytope.

- 6.15 Show that of the following two figures, the left is a Gale diagram for $C_4(8)$, while the second is the Gale diagram of a 2-neighborly 4-polytope with 8 vertices that is not cyclic (due to [531, p. 543]).



“Many” noncyclic neighborly polytopes can be constructed this way, by “small” modification of the Gale diagram of a cyclic polytope. (More were constructed by Shemer [494].)

- 6.16* Perles conjectured that every simplicial polytope is combinatorially equivalent to a face figure (iterated vertex figure) of an even-dimensional neighborly polytope. (Equivalently, every simple polytope is a face of a polar of a neighborly polytope.)

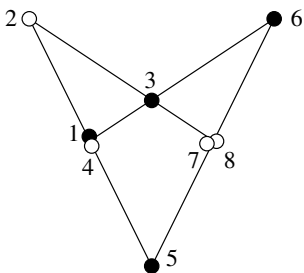
- (i) Define that a (finite) configuration of vectors in $\mathbb{R}^3$ is *uniform* if any 3 of the vectors span $\mathbb{R}^3$. Say that it is *balanced* if for every plane spanned by two vectors, the number of vectors on the two sides are equal.

Show that for simplicial d -polytopes with $d + 4$ vertices, the Gale diagram construction reduces Perles’ problem to the “embedding problem” of whether every uniform configuration of $d + 4$ vectors in $\mathbb{R}^3$ can be extended to a uniform balanced configuration.

- (ii) Using part (i), solve Perles’ problem for simplicial d -polytopes with $n = d + 4$ vertices: every simplicial d -polytope with $d + 4$ vertices is a quotient (i.e., combinatorially equivalent to an iterated vertex figure) of a neighborly $(2d + 4)$ -polytope with $2d + 8$ vertices.

(Hint: For part (i) you will need the characterization of Exercise 6.14. The Gale diagram formulation of Perles’ problem is due to Sturmfels. See Sturmfels [532, Sect. 7], where some partial results are also derived. Part (ii) was done by Kortenkamp [341].)

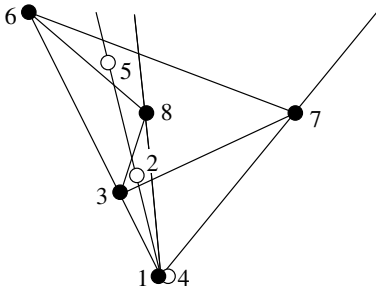
6.17 Show that the figure



is an affine Gale diagram of a 4-polytope P with 8 vertices, and verify the following facts. The polytope has 9 facets, four tetrahedra, four square pyramids, and an octahedron 235678. Every Gale diagram G' with the same set of positive circuits has 7 and 8 on the same point: so for every combinatorially equivalent polytope to P the vertices 2356 of the octahedron facet 235678 are coplanar. Thus the shape of the octahedron facet cannot be prescribed. However, show that the oriented matroid of P is not rigid.

6.18 Show that the figure below is the affine Gale diagram of a 4-polytope with 8 vertices and 11 facets, the *Kleinschmidt polytope* $K_4(8)$, and verify the following facts.

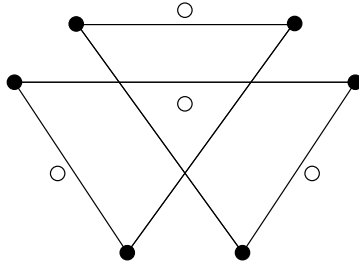
Except for the facet 235678, which is an octahedron, the facets of $K_4(8)$ are tetrahedra. The octahedron facet is not regular. No Gale diagram G' with the same set of positive circuits can have 136, 125, and 178 on lines: so there is no combinatorially equivalent polytope to $K_4(8)$ such that the octahedron facet is regular.



Show that the oriented matroid of $K_4(8)$ is not rigid. (Kleinschmidt [332] [203], Sturmfels [532, Fig. 6(a)])

6.19 (i) Construct a simple polytope for which the shape of a facet cannot be prescribed. For example, you might examine the polar of

the simplicial 6-polytope with 10 vertices, given by



(ii)* Can you prescribe the shape of a facet for simple 4-polytopes?

- 6.20 Let a centrally symmetric polytope with $2n$ vertices in $\mathbb{R}^d$ be given as $P = \text{conv}(X)$ for

$$X = \{\mathbf{u} \pm \mathbf{v}_1, \mathbf{u} \pm \mathbf{v}_2, \dots, \mathbf{u} \pm \mathbf{v}_n\}.$$

Show that the dependences and the value vectors of X can be reconstructed from those of the following set of only $n + 1$ points in $\mathbb{R}^d$:

$$X_0 = \{\mathbf{u}, \mathbf{u} + \mathbf{v}_1, \mathbf{u} + \mathbf{v}_2, \dots, \mathbf{u} + \mathbf{v}_n\}.$$

Thus the combinatorics of P can be read off from the dual configuration $G_0 \subseteq (\mathbb{R}^{n-d})^*$ to V_0 , the *central (Gale) diagram* of P , due to McMullen & Shephard [402].

Use central diagrams to classify the centrally symmetric polytopes with at most $2d + 2$ vertices for small values of d . (The case $d = 4$ was done by Grünbaum [252, Sect. 6.4].)

Argue that a complete classification of centrally symmetric polytopes with at most $2d + 2$ vertices is out of reach.

Show that the metric properties of this diagram are important, and that no further reduction to affine diagrams is possible.

Similarly, show that the signed circuits of X_0 do not determine those of X , so we cannot simply reduce to oriented matroid data [402].

- 6.21 Apply the Lawrence construction to three points on a line, either all three distinct, or two coinciding. What polytopes do you obtain? List all circuits and cocircuits both for the original configuration and for the Lawrence lifting.
- 6.22 Consider the prisms over simplices, $\text{prism}(\Delta_d)$, and construct their Gale diagrams. Show that they all arise as Lawrence polytopes.
- 6.23 Show that the oriented matroid given as an example for a 5-polytope with a nonprescribable 2-face is not rigid. (Use the fact that one can perturb the point 1 without changing any positive circuit.)
- Is Perles' example of a nonrational 8-polytope rigid?

6.24 A polytope is *projectively unique* if any combinatorially equivalent polytope can be obtained by a projective transformation.

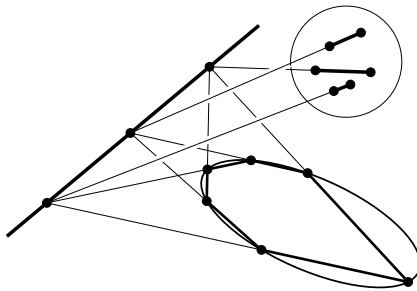
- (i) Show that in the plane, triangles and quadrilaterals are projectively unique, but n -gons with $n \geq 5$ are not projectively unique.
- (ii) Show that, more generally, d -polytopes with $n \leq d + 2$ vertices are projectively unique.
- (iii) Show that if P is projectively unique, then so is P^Δ . Conclude that d -polytopes with $n \leq d + 2$ facets are projectively unique.
- (iv) Derive from part (iii) that 3-polytopes with $f_1 \leq 9$ edges are projectively unique. (Use Euler's formula $v - e + f = 2$ from Exercise 4.6 or Lecture 8.)
Prove the converse, too.
- (v) Show that if P is projectively unique, and F is a face of P , then the face need not be projectively unique. In this situation, the face F cannot be arbitrarily prescribed.

6.25 How many (positive and negative) points do you need to create the Gale diagram of a rigid polytope from Suvorov's configuration?

6.26* What is the smallest number of points in a planar point configuration that violates the isotopy conjecture?

(At the moment the smallest known configurations have 14 points; see Example 6.24. In contrast, isotopy is known for $n \leq 9$ points, but only in the case of general position configurations, by Richter [455] [96, Sect. 8.2].)

6.27 Show that the polytope P of Example 6.23 can be constructed, via three Lawrence extensions, from the configuration of Pascal's Theorem ("the vertices of a hexagon lie on an ellipse if and only if the three intersection points of opposite sides are colinear"). This is indicated in the following picture.



Deduce from this construction method that this is a 5-polytope for which the shape of a 2-face cannot be prescribed.

(See Richter-Gebert [459, Example 3.4.3] for a detailed discussion.)

7

Fans, Arrangements, Zonotopes, and Tilings

Zonotopes are the images of n -cubes under affine projection maps. Since for most aspects of polytope theory n -cubes are not very complicated, this definition may hide the complexity and richness of this concept. Zonotopes are interesting from various points of view. Their combinatorial structure is closely linked to (and in a precise sense equivalent to) that of real linear hyperplane arrangements.

The aim of this lecture is to provide basic geometric intuition and the tools for a combinatorial description of zonotopes. We will see how zonotopes again are modeled by oriented matroids, and discuss the surprising appearance of general (“nonrealizable”) oriented matroids in the study of zonotopal tilings, and of hyperplane arrangements.

7.1 Fans

Definition 7.1. A *fan* in $\mathbb{R}^d$ is a family

$$\mathcal{F} = \{C_1, C_2, \dots, C_N\}$$

of nonempty polyhedral cones, with the following two properties:

- (i) Every nonempty face of a cone in $\mathcal{F}$ is also a cone in $\mathcal{F}$.
- (ii) The intersection of any two cones in $\mathcal{F}$ is a face of both.

The fan $\mathcal{F}$ is *complete* if the *union* $\bigcup \mathcal{F} := C_1 \cup \dots \cup C_N$ of $\mathcal{F}$ is $\mathbb{R}^d$, that is, if $\bigcup_{i=1}^N C_i = \mathbb{R}^d$. We will consider only complete fans here, and thus we will omit the word “complete” most of the time.